

# Homework 5

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## Exercise 1

### Part (a)

Since  $29 \bmod 13 = 3$ :

$$29^{202} \equiv 3^{202} \bmod 13$$

Since  $202 \bmod 12 = 10$ , by Fermat's Little Theorem:

$$\begin{aligned} 29^{202} &\equiv 3^{10} \bmod 13 \\ &\equiv 3 \bmod 13 \end{aligned}$$

### Part (b)

Since  $79 \bmod 9 = 7$ :

$$79^{79} \equiv 7^{79} \bmod 9$$

Since  $79 \bmod 8 = 7$ , by Fermat's Little Theorem:

$$\begin{aligned} 79^{79} &\equiv 7^7 \bmod 9 \\ &\equiv 7 \bmod 9 \end{aligned}$$

### Part (c)

Since  $99 \bmod 26 = 21$ :

$$99^{99999} \equiv 21^{99999} \bmod 26$$

Since  $26 = 2 \cdot 13$ ,  $\phi(26) = 12$ . Then, since  $99999 \bmod 12 = 3$ , by Euler's Theorem:

$$\begin{aligned} 99^{99999} &\equiv 21^3 \bmod 26 \\ &\equiv 5 \bmod 26 \end{aligned}$$

## Exercise 2

Clearly,  $645 = 3 \cdot 5 \cdot 43$  is composite. Furthermore, since  $644 \bmod 2 = 0$ :

$$\begin{aligned} 2^{644} &\equiv 2^0 \bmod 3 \\ &\equiv 1 \bmod 3 \end{aligned}$$

Then, since  $644 \bmod 4 = 0$ :

$$\begin{aligned} 2^{644} &\equiv 2^0 \bmod 5 \\ &\equiv 1 \bmod 5 \end{aligned}$$

Then, since  $644 \bmod 42 = 14$ :

$$\begin{aligned} 2^{644} &\equiv 2^{14} \bmod 43 \\ &\equiv 1 \bmod 43 \end{aligned}$$

By the Chinese Remainder Theorem,  $2^{644} \equiv 1 \bmod 645$ , so 645 is a pseudoprime to the base 2.

### Exercise 3

Clearly,  $2821 = 7 \cdot 13 \cdot 31$  is composite. We seek to show that  $a^{2821} \equiv a \pmod{2821}$  for all  $a$ ; by the Chinese Remainder Theorem, it is sufficient to prove the congruence modulo 7, 13, and 31. Then, since  $2821 \pmod{6} = 1$ :

$$\begin{aligned} a^{2821} &\equiv a^1 \pmod{7} \\ &\equiv a \pmod{7} \end{aligned}$$

Then, since  $2821 \pmod{12} = 1$ :

$$\begin{aligned} a^{2821} &\equiv a^1 \pmod{13} \\ &\equiv a \pmod{13} \end{aligned}$$

Then, since  $2821 \pmod{30} = 1$ :

$$\begin{aligned} a^{2821} &\equiv a^1 \pmod{31} \\ &\equiv a \pmod{31} \end{aligned}$$

Consequently, 2821 is a Carmichael number.

## Exercise 4

Let  $p, q$  be prime. Note that  $(p, q) = 1$ , and that  $p - 1, q - 1 \geq 1$ . By Fermat's Little Theorem,  $p^{q-1} \equiv 1 \pmod{q}$  and  $q^{p-1} \equiv 1 \pmod{p}$ . Then:

$$\begin{aligned} p^{q-1} + q^{p-1} &\equiv p^{q-1} \pmod{q} \\ &\equiv 1 \pmod{q} \end{aligned}$$

Similarly:

$$\begin{aligned} p^{q-1} + q^{p-1} &\equiv q^{p-1} \pmod{p} \\ &\equiv 1 \pmod{p} \end{aligned}$$

Consequently:

$$p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$$

## Exercise 5

Let  $p$  be prime. Suppose  $2^p - 1$  is composite. By Fermat's Little Theorem,  $2^p \equiv 2 \pmod{p}$ , so  $2^p - 2 = np$  for some integer  $n$ . Then:

$$\begin{aligned} 2^{(2^p-1)-1} &\equiv 2^{2^p-2} \pmod{2^p-1} \\ &\equiv 2^{np} \pmod{2^p-1} \end{aligned}$$

Further, since  $2^p \equiv 1 \pmod{2^p-1}$ :

$$\begin{aligned} 2^{(2^p-1)-1} &\equiv (2^p)^n \pmod{2^p-1} \\ &\equiv 1^n \pmod{2^p-1} \\ &\equiv 1 \pmod{2^p-1} \end{aligned}$$

Therefore  $2^p - 1$  is pseudoprime to the base 2, and so  $2^p - 1$  is either prime or pseudoprime to the base 2.

## Exercise 6

Let  $n$  be an integer such that  $3 \nmid n$ . Since  $63 = 7 \cdot 9$ , to show that  $n^7 \equiv n \pmod{63}$ , by the Chinese Remainder Theorem, it is sufficient to show that the congruence holds modulo 7 and 9. Suppose  $n \equiv 0 \pmod{7}$ ; then clearly  $0^7 \equiv 0 \pmod{63}$ . Otherwise, if  $n \not\equiv 0 \pmod{7}$ , note that  $(n, 7) = 1$ . Then, by Fermat's Little Theorem,  $n^7 \equiv n \pmod{7}$ . In either case,  $n^7 \equiv n \pmod{7}$ . Now note that  $\phi(9) = 6$ . Then, by Euler's Theorem,  $n^6 \equiv 1 \pmod{9}$ , so  $n^7 \equiv n \pmod{9}$ . Consequently,  $n^7 \equiv n \pmod{63}$  for all  $n$ .